Melooha Engineering Challenge — Final Submission Report

Theme: 'Help more people discover Melooha, trust it, and come back to it.'

Item	Details & Live Production URLs
Live Web App	**https://astro-psi-olive.vercel.app** (Mobile-First SPA on Vercel)
Live Backend API	**https://astro-5qqa.onrender.com** (FastAPI Docker on Render)
GitHub Repository	**https://github.com/AkshatRabgotra/Astro** (Public, CI passing)
Primary Metric	**Day-7 (D7) Retention & Weekly Active Return Rate (WARR)**
Core Retention Engine	**Personalized Muhurta Advisor (Google Calendar Sync) + Daily Habit Streak Economy**

1. Exploration & Opportunity Analysis

The First-Time User Experience in Melooha: Exploring the app reveals an enchanting cosmic onboarding, but also reveals the primary drop-off: *Birth Time Anxiety* and the *Post-Question Churn Cliff*. Once a user's single free question is spent, a standard payment paywall appears. For a curious user, a one-off birth chart reading feels like 'Static Fate'—there is no organic trigger to reopen the app tomorrow.

Competitor Teardown & Market Gaps:

App	Observed Model & Strengths	Critical Flaws / Why Users Churn	Melooha AI Advantage
AstroTalk	Human astrologer marketplace; large call volume.	**Predatory monetization**: ■30-100/min anxiety billing, aggressive gemstone/pooja upselling.	**Calm, private AI**: Zero fear-mongering, instant responses, complete privacy.
AstroSage	Comprehensive Vedic mathematical tables.	**1990s cluttered UI**: Overwhelmed with banner ads, dense astronomical jargon, no chat.	**Serene Cosmic UI**: Translates complex math into empathetic, human guidance.
Astroyogi	Curated human consultant directory.	**High friction**: Appointment scheduling delays, high per-session fees.	**24/7 Availability**: Instant sub-second streaming answers directly to your phone.
Kundali GPT	Free-form conversational natural language.	**Severe AI hallucinations**: Invents non-existent dashas and planetary degrees.	**In-Code Guardrails**: Swiss Ephemeris C-engine + Python AST verification.
Naksh	Minimalist modern interface.	**Lacks daily utility**: Static daily horoscope; no action-oriented tools.	**Actionable Muhurta**: 1-click Google Calendar sync for daily life decisions.

2. The Idea: From 'Static Fate' to 'Daily Decision Utility'

Target Persona: Ambitious professionals and seekers (21–42) wanting actionable daily timing for career, wealth, and relationships.
The One Metric It Moves: Day-7 (D7) Retention & Weekly Active Return Rate (WARR).
Why It Works: By transitioning astrology from a yearly static novelty into a **daily decision planner**, users receive tangible daily value.

The Three Core Value Engines:

• **Auspicious Muhurta Advisor (1-Click Google Calendar Sync):** Computes 30-min window sweeps combining Choghadiya, Tithi, Nakshatra, and Tara Bala personalized to the user's Moon. Directly adds auspicious windows to Google Calendar.

• **Daily Habit Streak & Rewards Economy:** Consecutive daily check-in rewards (+1 question credit, daily mantra). Eliminates dead-end paywalls and drives habitual daily app opens.

• **Proof-First Trust (Past Check & Dasha Twin):** Tests the user's chart against known past milestones (career changes, relocations) before predicting the future, creating unshakeable confidence in chart accuracy.

3. System Architecture & Technical Implementation

• **Swiss Ephemeris Core (pysweph):** High-precision C astronomical calculation engine executing within Docker in <15ms.
• **Multi-Provider Resilient LLM Orchestrator:** Primary Google Gemini 1.5 Flash $\rightarrow$ secondary Groq Llama-3.3-70b circuit breaker $\rightarrow$ offline deterministic engine. The user never sees a raw error.
• **In-Code Grounding Validator:** Mathematical fact-guardrail in Python that checks generated claims against computed ephemeris placements before streaming.
• **Neon Serverless PostgreSQL:** Production database with double-entry question balance ledger and event telemetry.
• **Vercel API Reverse Proxy:** Seamlessly proxies /api/* calls to Render Docker container, eliminating CORS hurdles.

4. Standout Highlights (Challenge Section 4)

• **Background Jobs & Calendar Sync:** Generates RFC-5545 calendar alerts and idempotent streak processing.
• **Automated Evaluation Harness:** 74 unit and integration tests passing in CI (ephemeris, grounding, balance flow).
• **Living Design System & Bilingual Voice:** Authentic Melooha dark-sky theme with complete English and Hindi support.

5. One-Command Local Startup

Run locally in a single command: .\run.ps1 (Windows) or make run (macOS/Linux).